Choppa Praveen Nooka Vinay Kumar

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LinkedIn — GitHub — LeetCode — Portfolio

Summary

Computer Science undergraduate at IIIT Kottayam with software engineering internship experience and hands-on backend development experience using Java and Spring Boot. Experienced in building REST APIs, implementing JWT authentication and authorization, designing relational data models, and handling transactional database operations. Developing a student buy-and-exchange platform with a focus on concurrency control, data consistency, and real-time communication. Interested in backend engineering and distributed systems.

Education

Indian Institute of Information Technology Kottayam Aug 2024 – Apr 2028

Bachelor of Technology in Computer Science and Engineering CGPA: 7.77

Experience

Software Engineering Intern Jun 2026 – Aug 2026

Lumenor AI Tech Pvt Ltd (Remote)

- Designed and developed **LumenDesignX**, a 3D rooftop solar design studio, translating product requirements into an end-to-end engineering workflow.
- Owned feature development across requirement analysis, technical design, implementation, debugging, and iterative improvements.
- Implemented solar design workflows with emphasis on modular architecture, state management, data consistency, and product integration.
- Collaborated with the engineering team on technical decisions, requirement refinement, and feature stability throughout the software development lifecycle.

Technical Skills

Languages: Java, SQL, Python, C

Backend Development: Spring Boot, Spring Security, REST API Development, JWT Authentication, Hibernate

Databases: PostgreSQL, MySQL, Relational Database Design

Backend Concepts: Object-Oriented Programming, Layered Architecture, DTOs, Authentication and Authorization, Database Transactions, Pessimistic Locking, Concurrency Control, Data Consistency

Tools: Git, GitHub, Maven, Postman, Thunder Client, Cloudinary

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks

Projects

NexCart – Student Buy-and-Exchange Platform GitHub

Java, Spring Boot, Spring Security, JWT, PostgreSQL, JPA/Hibernate, WebSocket

- Developing a student-focused platform for buying and exchanging items, with a feature-oriented backend architecture and RESTful APIs.
- Implemented JWT-based authentication and role-based access control using Spring Security to protect backend resources.
- Designed transactional checkout and inventory updates using **@Transactional** and pessimistic database locking to coordinate concurrent stock modifications and prevent overselling.
- Implemented real-time messaging using WebSocket, with JWT-based STOMP connection authentication, conversation-level authorization, and database-backed message persistence.
- Applied layered architecture, DTO-based request handling, input validation, and JPA/Hibernate persistence to support maintainable backend development.

GreenLeaf Nursery Management System GitHub

Java, Spring Boot, Spring Security, JWT, MySQL, Cloudinary

- Developed a nursery management application supporting public catalog browsing and protected administrative operations.
- Implemented JWT authentication and role-based authorization using Spring Security.
- Built REST APIs using layered architecture, DTOs, and JPA/Hibernate for database-backed operations.
- Integrated Cloudinary for image uploads and cloud-based media management.

RoadWatch AI – Smart Mobility Infrastructure Monitoring GitHub

Python, Flask, PostgreSQL, REST APIs, Neon, Render, Vercel

- Contributed to backend development and relational database implementation for a road infrastructure monitoring platform.
- Designed and populated PostgreSQL data models for users, authorities, contractors, road projects, complaints, and maintenance records.
- Integrated the Flask backend with PostgreSQL to support REST API workflows for complaint and infrastructure management.

Problem Solving

- Practicing algorithmic problem-solving with a focus on data structures, problem-solving patterns, and efficient implementations.

Certification

Fundamentals of Java Programming – Board Infinity, Coursera Mar 2025

Verify Certificate